Pham Anh Cuong

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PROFILE

Machine learning researcher working on efficient and trustworthy adaptation of foundation models, with prior experience in continual learning, low-rank and parameter-efficient fine-tuning, federated learning, and optimization. Currently working on uncertainty-aware LoRA fine-tuning across multiple downstream tasks, with broader interests in misalignment under LLM adaptation.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE — Aug 2024 - May 2026
Master of Science in Machine Learning — **GPA:** 3.72/4.0 (∼94/100) [Transcript]

- Thesis: "Learning in the Null Space: Small Singular Values for Continual Learning".
- Supervisors: Prof. Samuel Horváth, and co-supervised by Prof. Praneeth Vepakomma.
- Selected courses: Mathematics for AI, Advanced ML, Probabilistic & Statistical Inference, Optimization.

VNU University of Engineering and Technology (VNU-UET), Vietnam — Aug 2019 - Jun 2023
Bachelor in Information Technology (Honors Program) — **GPA:** 3.71/4.0 (8.78/10) [Transcript]

- Graduated with High Distinction, top 5% of the CS Department.
- Thesis: "Building a Vietnamese image captioning dataset and developing a visual-text transformation model for generating image captions" (Thesis Grade: 9.5/10).
- Supervisors: Prof. Quang-Thuy Ha, Dr. Van-Quang Nguyen, MS. Thi-Hong Vuong.

PUBLICATIONS

Conference Papers

[**C.2**] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth: "Learning in the Null Space: Small Singular Values for Continual Learning" (Third Conference on Parsimony and Learning (CPAL), 2026. Oral Presentation).

[**C.1**] Quynh-Trang Pham Thi, **Anh-Cuong Pham,** Ngoc-Huyen Ngo and Duc-Trong Le: "Memory-Based Method using Prototype Augmentation for Continual Relation Extraction" (IEEE RIVF '22).

Preprints

[**P.3**] Farshed Abdukhakimov, **Cuong Anh Pham**, Samuel Horváth, Martin Takáč, Slavomir Hanzely: "Polyak Stepsize: Estimating Optimal Functional Values Without Parameters or Prior Knowledge" (arXiv:2508.17288).

[**P.2**] Quang P.M. Pham, Khoi T. N. Nguyen, Nhi H. Doan, **Cuong A. Pham**, Qinbo Sun, Weimin Qi, Kentaro Inui, Dezhen Song: "SmallPlan: Leverage Small Language Models for Sequential Path Planning with Simulation-Powered, LLM-Guided Distillation" (arXiv:2505.00831).

[**P.1**] **Anh-Cuong Pham**, Van-Quang Nguyen, Thi-Hong Vuong, Quang-Thuy Ha: "KTVIC: A Vietnamese Image Captioning Dataset on the Life Domain" (arXiv:2401.08100).

RESEARCH EXPERIENCE

MBZUAI, Abu Dhabi, UAE

Research Engineer — Jun 2026 - present

- **Topic:** Uncertainty Quantification with LoRA-based fine-tuning
  Developing a framework for uncertainty-aware Bayesian LoRA fine-tuning across multiple downstream LLM tasks in terms of predictive performance, calibration, and failure detection.
  Supervisors: Prof. Praneeth Vepakomma

Graduate Research Assistant — Oct 2024 - May 2026

- **Topic:** Continual learning and low-rank adaptation
  Developed an orthogonality-based low-rank update method that uses small singular value subspaces to reduce catastrophic forgetting while maintaining competitive model accuracy, resulting in a first-author oral paper [C.2] and currently expanding to continual fine-tuning.
  Supervisors: Prof. Samuel Horváth and Prof. Praneeth Vepakomma

- **Topic:** Parameter-free stochastic optimization
  Co-developed an adaptive Polyak-style stepsize method based on twin iterates that removes the need for prior knowledge of the optimal objective value and contributed to the experimental implementation, leading to preprint [P.3].
  Supervisors: Prof. Samuel Horváth and Prof. Martin Takáč
- **Topic:** Federated optimization and fine-tuning
  Reproduced and analyzed Fed-SB and LoRA-SB under continual learning and multi-task learning setups.
  Supervisors: Prof. Praneeth Vepakomma and Prof. Samuel Horváth
- **Topic:** Small language models for robotic path planning (extended from MBZUAI coursework project)
  Utilized SLMs with the guidance from LLMs to enable efficient and adaptive real-time inference on edge devices for autonomous robotics path-planning tasks, leading to preprint [P.2].

DSKT Laboratory at VNU-UET, Hanoi, Vietnam — Sep 2021 - Sep 2023
Student Research Intern

- Built a Vietnamese image-captioning dataset and evaluated it with a lightweight Transformer-based captioning model, leading to preprint [P.1].
- Contributed to a prototype-augmentation method for continual relation extraction, resulting in the paper [C.1].

INDUSTRY EXPERIENCE

Viettel Networks, Hanoi, Vietnam — Oct 2023 - Jul 2024
Data Scientist

- Developed computer-vision pipelines for telecommunication maintenance tasks, using segmentation, classification, and image-processing methods; resulted in an internal tool to support network maintenance.
- Built predictive and data-processing workflows over datasets ranging from 10M to 1B records to optimize network-resource allocation.

GHTK Joint Stock Company, Hanoi, Vietnam
R&D Engineer — Aug 2022 - Aug 2023

- Co-developed and deployed an internal email spam classifier, achieving an F1 score of 0.90 on a corpus of ∼50,000 emails.

R&D Engineer Intern — May 2022 - Aug 2022

- Developed a Vietnamese document classification prototype using doc2vec during the internship program.

TALKS & PRESENTATIONS

"**Learning in the Null Space: Small Singular Values for Continual Learning**"

- Oral and Poster Presentation at CPAL (Tübingen, Germany) — 2026
- At RelationalML Group at CISPA, hosted by Prof. Rebekka Burkholz (online) — 2026

HONORS & AWARDS

Full scholarship for 2 years of the Master of Science program at MBZUAI — 2024
Third Prize (5th place) in Linear Algebra, VNU-UET Mathematical Olympiad — 2023
Third Prize (6th place) in Linear Algebra, VNU-UET Mathematical Olympiad — 2022
Yamada Scholarship (6 students/year at VNU-UET) — 2022
Third Prize (5th place) in Linear Algebra, VNU-UET Mathematical Olympiad — 2021
Academic Excellent Scholarship at VNU-UET (5 selected students) — 2021
Selected to represent Nghe An province (top 10 students) in Vietnam Mathematical Olympiad — 2019

SKILLS

Programming: Python, C/C++
ML Frameworks: PyTorch, Scikit-learn, Transformers, PEFT.
Additional: LaTeX, Git.
Language: English (IELTS 7.0), Vietnamese (native).

REFERENCES

Dr. Samuel Horvath, Assistant Professor at MBZUAI (UAE): (samuel.horvath@mbzuai.ac.ae)
Dr. Praneeth Vepakomma, Assistant Professor at MBZUAI & Visiting Professor at MIT: (vepakom@mit.edu)